Yilin Liu

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Research Interests

Efficient AI systems; LLM inference and serving; low-precision computation; KV-cache quantization; GPU-aware system-algorithm co-design; agent systems.

Education

Hong Kong University of Science and Technology (Guangzhou) Sep. 2023 -- Jun. 2027 (expected)
B.Sc. in Data Science and Big Data Technology *Guangzhou, China*

- GPA: 3.692/4.300; coursework in machine learning, deep learning, reinforcement learning, and optimization.

University of California, Berkeley Aug. 2025 -- Dec. 2025
Berkeley Global Access Program, Visiting Student *Berkeley, CA, USA*

- GPA: 3.910/4.000; coursework in computer architecture, agentic AI, and data mining.

Research Experience

Efficient LLM Inference and Low-Precision KV-Cache Quantization May 2026 -- Present
Undergraduate Researcher, HKUST(GZ); advised by Prof. Zeyi Wen *Guangzhou, China*

- Study efficient autoregressive decoding with low-bit KV caches, focusing on the interaction among quantization algorithms, data layouts, and GPU execution paths.
- Analyze related systems including QServe, BitDecoding, TurboAttention, FlashInfer, and vLLM; design accuracy, latency, memory, and H100 profiling evaluations.

ClawMobile: Efficient and Reliable Mobile Agents Jun. 2026 -- Present
Undergraduate Research Intern, MBZUAI UGRIP *Online / Abu Dhabi, UAE*
Advised by Prof. Chun Jason Xue and Prof. Youcheng Sun.

- Investigated token efficiency and execution reliability for smartphone-native agents operating in real Android environments.
- Designed capability recipes and evaluation workflows for token usage and task completion; the team received the UGRIP 2026 Best Project Award.

Agentic Mathematical Data Synthesis Mar. 2025 -- Apr. 2026
Undergraduate Researcher; advised by Prof. Jiaheng Wei *Guangzhou, China*

- Contributed to a multi-agent framework for synthesizing compact, high-quality mathematical reasoning data for efficient LLM fine-tuning.

Publication

AgenticMath: Enhancing LLM Reasoning via Agentic-based Math Data Generation.
Xianyang Liu, **Yilin Liu**, Shuai Wang, Hao Cheng, Andrew Estornell, Yuzhi Zhao, Jun Shu, and Jiaheng Wei. DATA-FM Workshop at ICLR 2026. [Paper]

Honors and Awards

2026	Best Project Award, MBZUAI Undergraduate Research Internship Program (UGRIP)
2026	2025 INFH-Lizhi Innovative Talent Scholarship, HKUST(GZ)
2026	Outstanding Award, Ascend AI Innovation Competition - Operator Challenge, HKUST(GZ) Special Session
2025	Second Prize, Blue Bridge Cup National Software and IT Talent Competition, Guangdong Division
2024	Gold Medal and Best New Basic Part Nomination, International Genetically Engineered Machine Competition (iGEM)

Technical Skills

Programming: Python, C++, C, Bash, Git.
ML and systems: PyTorch, Hugging Face, Triton (learning), CUDA profiling, Nsight Compute, FlashInfer, vLLM, model evaluation, quantization.
Research: Literature review, experimental design, benchmarking, data analysis, technical writing, and presentation.